

2025 AMC 10A Problem 16 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 16

**Problem:**

There are three jars. Each of three coins is placed in one of the three jars, chosen at random and independently of the placements of the other coins. What is the expected number of coins in a jar with the most coins?

Answer Choices:

A. $\frac{4}{3}$

B. $\frac{13}{9}$

C. $\frac{5}{3}$

D. $\frac{17}{9}$

E. 2

Solution:

We have 3 coins and 3 jars. Each coin is placed independently and uniformly at random into one of the 3 jars.

Hence, there are $3^3 = 27$ possible outcomes, all equally likely.

Let M denote the number of coins in the jar with the most coins.

Each possible arrangement of the 3 coins among 3 jars must result in one of the following occupancy types:

Type	Description	Maximum occupancy
$(3, 0, 0)$	All 3 coins in one jar	3
$(2, 1, 0)$	Two in one jar, one in another	2
$(1, 1, 1)$	One in each jar	1

We can also calculate the number of possible assignments of each type:

Type $(3, 0, 0)$: 3 ways (choose which jar gets all 3 coins).

Type $(2, 1, 0)$: $\binom{3}{2}$ (choose the 2 coins together) $\times$ 3 (choose jar for them) $\times$ 2 (choose jar for single coin) $= 3 \times 3 \times 2 = 18$.

Type $(1, 1, 1)$: $3! = 6$ ways (each coin in a distinct jar).

We finish by calculating the expected value of M :

$$\begin{aligned}
 \mathbb{E}[M] &= \frac{3 \cdot 3 + 18 \cdot 2 + 6 \cdot 1}{27} \\
 &= \frac{9 + 36 + 6}{27} \\
 &= \frac{51}{27} = \boxed{(D) \frac{17}{9}}.
 \end{aligned}$$

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